

Chapter 10

Late Complications of Radiation Therapy in Uveal Melanoma



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Introduction

Since reports from the Collaborative Ocular Melanoma Study (COMS), radiation therapy has become a primary treatment for uveal melanoma [13, 19]. It offers a favorable alternative to enucleation as patients are often able to maintain some visual acuity in their affected eye following treatment. Nonetheless, although plaque brachytherapy has similar survival rates compared to enucleation for small- and medium-sized tumors [19], radiation therapy results in significant side effects. Although some level of visual acuity may remain in the affected eye, in the years following radiation, visual side effects may be experienced by patients to various degrees, due to radiation damage to important ocular structures such as the iris, lens, optic nerve, and retina. Early studies in the treatment of ocular tumors demonstrated severe visual impairments in up to 50% of eyes in the years following radiation

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therapy; often, these visual side effects are not observed until several months or years after treatment.

The most commonly seen ocular complications from radiation therapy are cataracts, radiation retinopathy, and radiation optic neuropathy [35]. Five years after radiation, the COMS trial reported 83% of study eyes with cataract formation [15], and Gunduz et al. reported an estimated 42% and 8% of patients with nonproliferative and proliferative radiation retinopathy, respectively [36]. A retrospective study by Kinyoun et al. reported development of optic neuropathy in 55% of study eyes with a median follow-up time of 28 months [47]. Though these complications may be seen in any patient treated with radiation, the extent and effect of radiation damage depends heavily on the size and location of the tumor as well as the specifications of the radiation treatment itself.

Radiation Retinopathy

Radiation retinopathy, defined as changes to the microvasculature in the months or years following radiation treatment for ocular or head/neck cancers, is one of the most common side effects of radiation treatment. For a large proportion of patients, radiation retinopathy is a slowly progressive disease that can cause devastating and sometimes irreversible damage to their visual acuity. There are several risk factors that physicians should be aware of when planning radiation treatment and follow-up visits. Fortunately, there are promising treatments currently being studied to improve visual acuity damage and to potentially prevent damage from retinopathy.

Etiology and Clinical Features

During radiation treatment, the retinal vascular endothelial cells may be damaged, leading to a devastating domino effect on the retinal vasculature. The damage can result in one of the most common visual side effects of radiation therapy – radiation retinopathy. Radiation retinopathy, also known as radiation maculopathy, generally presents between 6 months and 3 years following treatment in a significant portion of patients [13]. In a previous study of patients treated for melanoma with plaque brachytherapy, 30% of patients presented with evidence of radiation retinopathy within 2 years of treatment [50]. Other studies have reported various rates of disease development, though it is generally understood that the risk of radiation retinopathy can rely heavily on type of radiation, fractionation schedule, tumor characteristics, and radiation dosage.

Similar to other diseases of the retinal vasculature, such as diabetic retinopathy, radiation retinopathy can present in two stages: nonproliferative and proliferative [48]. Most patients affected are classified as nonproliferative retinopathy in the

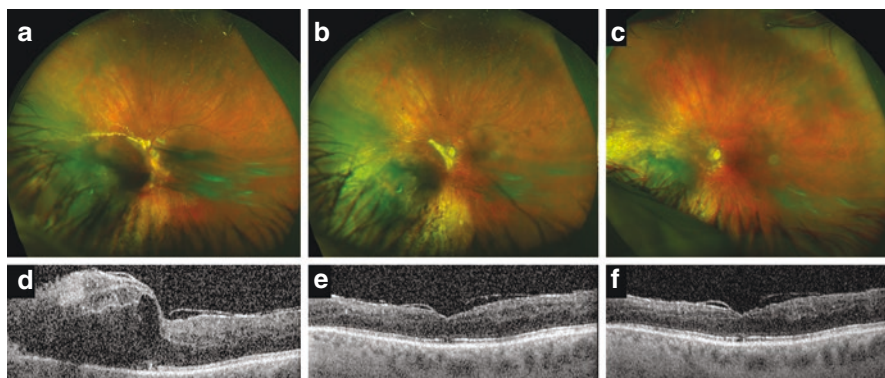


Fig. 10.1 Presentation and treatment of radiation retinopathy following treatment for choroidal melanoma. A 50-year-old male presented with a choroidal melanoma posterior to the equator in the left eye and was treated with Iodine-125 plaque brachytherapy. (a and d) Nineteen months later, the patient presented with cystoid macular edema, intraretinal exudates, retinal hemorrhage, and RPE atrophy. The patient then initiated treatment with monthly intravitreal injections of ranibizumab. (b and e) At 6 months, the patient presented with improved CME, improved intraretinal exudates, and resolved hemorrhage. (c and f) At 12 months, the patient presented with resolved CME, resolved exudates, and resolved hemorrhage

early stages of the disease; advancement to a proliferative classification occurs when growth factors are upregulated and induce neovascularization in the retina, a progression that occurs in a small portion of severe cases [7]. Upon presentation, clinical features of radiation retinopathy include intraretinal hard exudates, cotton wool spots, macular edema, microaneurysms, and retinal ischemia (Fig. 10.1) [21]. Atrophy of the retinal pigment epithelial (RPE) cells may also be observed.

Risk Factors

Physicians should be aware of several risk factors for radiation retinopathy when treating patients with radiation therapy. Risk factors can range from radiation-level factors to patient demographics (Table 10.1). In the years following radiation therapy, patients should be cognizant of any visual changes they experience; patients should also be regularly assessed by an ophthalmologist to rule out any early anatomical changes indicative of radiation retinopathy.

Types of Radiation

Although late complications from radiation have been observed across all types of radiation used to treat uveal melanoma (i.e., external beam radiation, plaque brachytherapy, proton beam radiation, gamma knife), incidence and extent of visual side

Table 10.1 Risk factors for radiation retinopathy

Risk factor	Details
Types of radiation	Gamma knife radiation – in severe VA loss in nearly all eyes External beam radiation and plaque brachytherapy – VA loss in approximately 40–60% of eyes
Radioactive isotopes	Cobalt-60 – high rates of retinopathy due to high energy gamma emission/poor shielding Ruthenium-106 and Strontium-90 – beta emitters, less VA loss due to rapid dose fall-off Iodine-125 and Palladium-106 – low-energy photon emitters. Palladium has a more rapid dose fall-off and thus may result in lower rates of radiation retinopathy compared to iodine
Dosimetry and fractionation	<70 Gy to the tumor and <40 Gy to the retina are associated with lower rates of retinopathy If possible, hyperfractionation can lower the risk of radiation damage
Tumor characteristics	Larger tumors require higher doses of radiation and are associated with higher incidence of retinopathy Proximity to the macula is associated with worse radiation damage
Patient demographics	Diseases of the vasculature such as diabetes, hypertension, or coronary artery disease increase risk of retinopathy

effects may differ between treatment modality. In the COMS study, among eyes with medium-sized tumors treated with plaque brachytherapy, 66% and 45% of eyes had a visual acuity worse than 20/40 and 20/200 3 years after radiation treatment, respectively [13]. A study by Gunduz et al. reported radiation retinopathy in 42% of eyes treated with plaque brachytherapy 5 years after treatment [36]. Among eyes treated with proton beam radiation for parapapillary choroidal melanoma, 68% of patients developed radiation papillopathy a mean of 1.5 years after treatment; 42% of these eyes, however, retained useful vision and a third spontaneously recovered vision over 5 years [45]. By far the modality with the highest rates of visual damage is gamma knife radiation. In a study of patients treated with gamma knife radiosurgery for choroidal melanoma, visual acuity loss was severe and highly prevalent with only one of 32 patients (3%) retaining vision of 20/40 or better after a median 38 months of follow-up [38].

Radioactive Isotopes

Type of radioactive isotope also plays a role in the amount of radiation damage to the eye. Several different isotopes have been studied and utilized in plaque brachytherapy over the last several decades with differing risks of radiation damage compared to others. In 1948, Stallard began treating uveal melanoma with Cobalt-60 radiation applicators, a technique that was used through the 1970s and 1980s for choroidal melanoma [80]. However, the popularity of Cobalt-60 fell out due to its high-energy gamma emission which made it challenging to shield from other important ocular structures and OR staff [76]. Today, Iodine-125, extensively studied in

the COMS trials [11], is one of the most popular radioactive isotopes used in plaque brachytherapy, especially in the United States. Iodine-125 emits low-energy photon radiation and has excellent tumor control and mortality rates when compared to enucleation [73].

Ruthenium-106 and Strontium-90 have also been studied for their effectiveness in treating ocular cancers, and Ruthenium-106 is commonly used in plaque brachytherapy in Europe. Both isotopes are classified as beta emitters and are known to have rapid dose fall-off which can help minimize unnecessary doses of radiation to adjacent ocular structures compared to other isotopes [62]. However, the use of Ruthenium-106 generally delivers a high dose of radiation to the sclera in order to fully penetrate the tumor and thus is not often used for tumors over 5 mm in apical height. Some have also reported inferior local tumor control rates when compared to Iodine-125, likely due to a lack of adequate radiation to the tumor apex beyond the area of radiation fall-off [6, 54, 70, 87]. Similar mortality rates have been reported when comparing treatment of thin tumors [54]. Despite this restriction, studies in Ruthenium-106 with properly chosen cases have demonstrated excellent long-term results for patients treated for ocular tumors with favorable results in visual acuity. In a large study by Bergman et al., a visual acuity of 20/200 or worse was observed in only 38% of patients at 3 years compared to 49% treated with Iodine-125 in the COMS study, with comparable heights of tumors between studies [6]. Thus, some physicians may opt for treatment with Ruthenium-106 for thinner tumors when given the option between isotopes.

Research has also been conducted for Palladium-106, categorized as low-energy photon radiation with a more rapid radiation dose fall-off compared to Iodine-125 [62]. It has been proposed as an alternative to Iodine-125 with the benefit of lower radiation doses to surrounding ocular structures. Additionally, Palladium-106 seeds are similar in size to Iodine-125 and thus can be used in identical plaques. Limited research has been conducted with this isotope, though lower rates of radiation retinopathy compared to Iodine-125 have been observed in single-center reports in the published literature [25]. Nevertheless, investigators must be wary that the rapid dose fall-off may result in increased rates of local treatment failure.

Radiation Dosimetry and Fractionation

Radiation dosimetry and fractionation are known to be important risk factors for late complications of radiation treatment. Recent advancements in technology have allowed radiation therapists to plan and calculate dosimetry to certain parts of the tumor and to surrounding structures prior to radiation treatment. Generally, lower total doses to the tumor (<70 Gy) and to the retina (<40 Gy) are associated with lower rates of radiation retinopathy and radiation complications. If possible, hyperfractionation of radiation doses with lower doses at each session can also lower the risk of radiation damage [60, 65]; for modalities such as plaque brachytherapy, however, hyperfractionation is not possible. Total dose reduction to an apex dose

less than that reported in the COMS is controversial and has been explored in non-randomized studies [66].

Tumor Characteristics

Several studies have reported on the association between specific tumor characteristics and rates of radiation complications as well. Tumors that are larger at baseline typically require higher doses of radiation for treatment and are thus associated with a higher incidence of radiation retinopathy and other complications [12, 14, 42]. Tumor location at baseline is also critical when assessing risk of radiation damage to other important ocular structures such as the retina, optic nerve, lens, sclera, and other structures. For example, for peripapillary tumors, the risk of optic disc edema or optic neuropathy is increased compared to equatorial tumors.

Patient Demographics

Patient-level factors can increase the risk of radiation retinopathy significantly when treated with radiation therapy. Patients with diseases of the vasculature such as diabetes, hypertension, or coronary artery disease [64, 85, 86] are already at risk for retinopathies and have increased incidence when treated with radiation to the eye.

Treatment for Radiation Retinopathy

Though radiation retinopathy has been a well-established side effect of radiation treatment for decades, there is no widely accepted management strategy for eyes affected by the disease. However, due to the clinical similarities to other exudative retinal diseases such as diabetic retinopathy and neovascular age-related macular degeneration, multiple similar treatment techniques have been studied. Although there is no treatment currently approved by the US Food and Drug Administration, studies in laser photocoagulation and intravitreal pharmacotherapy have produced promising results for patients affected by radiation retinopathy.

Laser Photocoagulation

Early research in the treatment of radiation retinopathy explored the efficacy of various laser photocoagulation techniques to resolve anatomical changes and improve visual acuity in patients. Overall, improvements in anatomic and visual outcomes appeared to be modest and short-lived among these patients. In 1996, Kinyoun et al. reported a small retrospective case series in which 11 eyes with proliferative

radiation retinopathy were treated with panretinal photocoagulation [48]. Through a mean follow-up time of 75 months, neovascularization had resolved in 91% of treated eyes, though these eyes still demonstrated poor visual acuity following treatment. In a study by Hykin et al., focal laser for radiation-induced macular edema modestly improved visual acuity in the short term though the one-time treatment did not seem to sustain visual acuity improvements through 1 or 2 years of follow-up [41]. The use of photodynamic therapy (PDT) was reported by multiple authors with promising improvements in neovascularization, visual acuity, and anatomic outcomes, though most of these studies were small, retrospective case series with limited follow-up time [3, 4, 51].

Intravitreal Steroids

Intravitreal injections of corticosteroids such as triamcinolone acetonide and dexamethasone have been theorized to help improve visual and anatomic outcomes in patients with radiation retinopathy due to the ability of corticosteroids to restore capillary permeability and blood-retina barriers. However, research into this technique has remained limited to small case series and anecdotal reports with minimal follow-up. Two small, prospective case series investigated treatment of radiation-induced macular edema and radiation papillopathy with intravitreal injections of triamcinolone acetonide [77, 78]. Both reports demonstrated vast and meaningful improvements in visual acuity and central macular thickness (CMT) within the first weeks following treatment, though follow-up in these cases were extremely limited. A retrospective case series by Baillif et al. investigated dexamethasone implants to treat radiation-induced macular edema, with short-term improvements in visual acuity; visual improvements did not last past 5 months [2].

Multiple studies have also used intravitreal corticosteroids to treat recalcitrant radiation retinopathy, previously treated with off-label bevacizumab, a well-known anti-vascular endothelial growth factor (VEGF), and/or laser photocoagulation [5, 39, 44, 71, 81]. For the majority of patients, stabilization or improvement of visual acuity or CMT regressed within 6 months of treatment.

Intravitreal Anti-vascular Endothelial Growth Factors

In recent years, the advent of intravitreal anti-VEGF agents such as bevacizumab, ranibizumab, and aflibercept has revolutionized the treatment of exudative retinal diseases, including radiation retinopathy. Several retrospective case series and anecdotal reports have demonstrated the beneficial visual and anatomic effects of intravitreal bevacizumab and ranibizumab, and more recently, prospective studies have examined the use of aflibercept (Fig. 10.1) [23, 29, 30, 33, 37, 43, 55, 56, 61, 72, 74]. Similar to laser photocoagulation and corticosteroid treatment, discontinuation of anti-VEGF treatment appears to result in recurrence of symptoms of radiation retinopathy. Unlike other treatments, however, anti-VEGF injections can be given

frequently to maintain and improve visual and anatomic outcomes in patients affected by radiation retinopathy. Still, the challenge remains of how often to treat patients to minimize burden while maximizing the visual and anatomic gains of anti-VEGF treatment.

To date, the prospective literature in this field has been limited (Table 10.2). In 2013, Finger and Chin published a small, prospective, nonrandomized clinical trial of 10 eyes from patients with recalcitrant radiation retinopathy from external beam radiation or plaque brachytherapy [29]. Patients had previously been treated with anti-VEGF treatment with incomplete response, defined as persistent macular edema or leakage. Over the course of 1 year, subjects received 12 injections of high-dose ranibizumab (2.0 mg). Seventy percent of subjects completed the trial of which 100% had stable or improved BCVA by the year 1 endpoint. Thirty percent of subjects withdrew from the trial due to worsening retinopathy. By the end of the study, 80% of eyes also had statistically significant improvement of central foveal thickness.

Scheffler et al. reported the use of ranibizumab with or without targeted retinal photocoagulation (TRP) for radiation-induced macular edema in a phase IIb, randomized, prospective study [72]. Forty eyes from 40 subjects were randomized to one of three cohorts and treated with monthly or pro re nata (as needed, PRN) ranibizumab (0.5 mg) with or without TRP through 1 year. By the year 1 endpoint, 97% of eyes had VA 20/400 or better; the monthly cohort had the largest visual gains, and there was no indication of significant benefits of TRP. CMT gains were also significant among eyes treated with monthly ranibizumab, and anatomic improvements were significant among all three cohorts. In the second year of the study, all subjects were transitioned to a treat-and-extend protocol dependent on the exudative state of their macula at each visit. Results from year 2 are ongoing.

Aflibercept has also been a promising anti-VEGF agent for exudative retinal diseases and is now FDA approved for wet AMD and all stages of diabetic retinopathy. Its efficacy in the treatment of radiation retinopathy has also been documented through two prospective studies. In a study by Fallico et al., nine eyes from nine subjects with radiation-induced macular edema were treated with 2.0 mg aflibercept every 4 weeks until maximum visual acuity was achieved or until there were no signs of edema on optical coherence tomography; PRN treatment was enacted thereafter [23]. Overall, a mere 4.4 mean injections were given per eye over the 24-month study period; visual acuity and central foveal thickness also significantly improved by the end of study. Although the study was undoubtedly limited by a small number of patients, the results of the prospective case series were promising.

A larger, prospective, randomized trial reported by Murray et al. detailed two treatment regimens for radiation maculopathy – intravitreal injections of aflibercept every 6 weeks or treat-and-extend centered around every 6 weeks [61]. Through 1 year of follow-up, significant improvements in central retinal thickness were observed and 42.5% of subjects had VA 20/50 or better. There were no significant differences in injection frequency between the two cohorts.

Table 10.2 Prospective clinical trials in anti-vascular endothelial growth factor therapy

Author (year)	Subjects	Cohorts	Length of study	Outcomes by end of study
Finger et al. (2013)	Recalcitrant radiation maculopathy with a history of failed anti-VEGF treatment	<i>Cohort 1</i> (<i>n</i> = 5): high-dose (2.0 mg) ranibizumab every 30 days <i>Cohort 2</i> (<i>n</i> = 5): high-dose (2.0 mg) ranibizumab every 30 days for 4 months, PRN high-dose ranibizumab thereafter	12 months	Stable/improved BCVA in 70% of eyes Significant improvement in CFT in 80% of eyes
Kim et al. (2016)	Choroidal melanoma within 2 disc diameters of the optic nerve and/or macula with no history of retinopathy or anti-VEGF treatment	<i>Cohort 1</i> (<i>n</i> = 20): 0.5 mg prophylactic ranibizumab every 2 months <i>Cohort 2</i> (<i>n</i> = 20): 0.5 mg or 1.0 mg prophylactic ranibizumab every 2 months	22 months	VA 20/200 or better in 97% of eyes Evidence of radiation maculopathy in 33% of eyes
Fallico et al. (2019)	Treatment-naïve radiation maculopathy secondary to Ru-106 brachytherapy	2.0 mg aflibercept monthly for all subjects (<i>n</i> = 9) until maximum VA and/or a dry macula is achieved; PRN aflibercept thereafter	24 months	Mean 4.4 injections received BCVA significantly improved CFT significantly decreased
Murray et al. (2019)	Treatment-naïve radiation maculopathy from radiation therapy	<i>Cohort 1</i> (<i>n</i> = 20): 2.0 mg aflibercept every 6 weeks <i>Cohort 2</i> (<i>n</i> = 20): 2.0 mg aflibercept and treat-and-adjust thereafter (1 week adjustments)	60 weeks	BCVA of 20/200 or worse in 5% of eyes CRT significantly improved No significant differences in injection frequency between cohorts
Scheffler et al. (2019)	Treatment-naïve radiation-induced macular edema with a history of radiation therapy	<i>Cohort 1</i> (<i>n</i> = 8): 0.5 mg ranibizumab monthly for 12 months; treat-and-extend thereafter <i>Cohort 2</i> (<i>n</i> = 16): 0.5 mg ranibizumab monthly and TRP 1 week after first injection for 12 months; treat-and-extend thereafter <i>Cohort 3</i> (<i>n</i> = 16): 0.5 mg ranibizumab monthly and TRP 1 week after first injection for 3 loading doses, then PRN for 12 months; treat-and-extend thereafter	24 months	BCVA 20/400 or better in 97% of patients Most significant visual gains in Cohort 1 No visual or anatomic benefit with TRP treatment

VEGF vascular endothelial growth factor, PRN *pro re nata*, BCVA best-corrected visual acuity, VA visual acuity, CFT central foveal thickness, CRT central retinal thickness, TRP targeted retinal photocoagulation

All four studies in ranibizumab and aflibercept demonstrated promising visual and anatomic results for the treatment of radiation retinopathy or radiation-induced macular edema, though further efficacy and safety data will need to be confirmed in larger, multicenter trials.

Prophylactic Treatment

Several possible prophylactic treatments of radiation retinopathy have also been explored in retrospective case series and prospective trials. In prospective literature for other exudative retinal diseases, early treatment and prophylactic approaches appear to be visually and anatomically beneficial for patients [89]. For patients treated with radiation therapy for ocular or head/neck cancers, prophylactic treatment may be a reasonable approach to preserve long-term vision due to the long latency period between the end of radiation treatment and the onset of visual symptoms from retinopathy. Multiple studies have investigated the use of prophylactic laser photocoagulation and/or intravitreal corticosteroids at the time of radiation therapy [31, 40, 57], which appeared to reduce the risk of developing signs of retinopathy within the first 2 years following radiation treatment. A prospective case series by McCannel and McCannel used pars plana vitrectomy and silicone oil injection at the time of radiation treatment [58]. In theory, the oil would attenuate the excess radiation exposure to adjacent ocular structures. Through a mean 2 years of follow-up, subjects had lower rates of retinal hemorrhage, intraretinal exudates, edema, and other retinal changes compared to patients not treated with a silicone oil injection.

Some ophthalmologists have also explored the use of prophylactic intravitreal anti-VEGF injections beginning at the time of plaque removal. Two small retrospective studies analyzed the use of bevacizumab given every 4 months in the 2 years following radiation treatment [75, 79]. Overall, subjects appeared to have fewer long-term retinal complications and better visual acuity at 2 years post-plaque brachytherapy compared to patients not treated with prophylactic bevacizumab. In 2016, Kim et al. reported a prospective, phase I trial studying the efficacy of prophylactic anti-VEGF in patients receiving radiation treatment for choroidal melanoma [46]. Through 24 months, 40 eyes received 0.5 mg or 1.0 mg ranibizumab every 2 months. By the end of the study, 97% of eyes had a VA of 20/200 or better compared to 45% of historical controls, and 33% of eyes with a small/medium tumor at baseline had evidence of radiation retinopathy at 24 months compared to 68% of historical controls. No serious adverse events related to the study drug were observed throughout the course of the study.

The possibility of prophylactic treatment for radiation retinopathy may be promising for many patients who may experience anxiety at the thought of constantly monitoring their vision for decreases in acuity. Additionally, it may provide a standardized treatment protocol for patients following treatment of ocular cancers. However, before these techniques may be used in routine clinical practice, larger, multicenter trials must be conducted to determine efficacy and safety among patients.

Radiation Effects on Other Ocular Structures

Although damage to the retina is one of the most common visual side effects of radiation therapy, other important ocular structures may be damaged during treatment and require attention. Risk factors and therapeutic intervention for these side effects may vary, though proximity and quantity of radiation exposure plays a role in the incidence and extent of damage to all ocular structures.

Iris

Radiation treatment can cause direct radiation damage to anterior structures of the eye, resulting in thinning and atrophy of the iris [24, 26, 35]. This direct damage is more common when treating anterior uveal melanoma affecting the iris, the ciliary body, and/or the anterior choroid. Damage to the posterior portions of the eye may also result in indirect side effects on the iris. As discussed, radiation-induced ischemia in the retina results in an upregulation of VEGF molecules in the vitreous, leading to neovascularization of the retina and to neovascularization of the iris, specifically on the angle between the iris and the cornea [8, 27, 32, 82]. Similar to neovascularization in the retina, these new blood vessels in the iris are often weaker than normal, mature blood vessels and result in fluid leakage or inflammation. Eventually, the drainage angle can close, leading to increased intraocular pressure, neovascular glaucoma, pain, and abnormal vision.

Neovascularization of the iris can sometimes be treated in combination with treatment of posterior radiation damage. Although laser photocoagulation may be beneficial in the short term of reducing neovascularization of the iris, inhibition of VEGF agents has been shown to be one of the most effective treatments. Multiple anecdotal case reports and case series have demonstrated the use of anti-VEGF agents in treating neovascularization of the iris caused by various radiation treatments [9, 20, 63, 84, 91]. A few small, prospective clinical trials have also described the use of bevacizumab and ranibizumab in treating patients with neovascular glaucoma caused by other retinal diseases [53, 88, 90]. In these cases, anti-VEGF injections were able to reduce iris neovascularization and intraocular pressure through the study period.

Lens

Radiation cataracts are especially prevalent in patients treated with radiation therapy to the orbital. In a retrospective case series by Kinyoun, 52% of patients treated with either external beam irradiation or plaque brachytherapy for various indications demonstrated development of new cataracts [47]. By the 5-year endpoint in

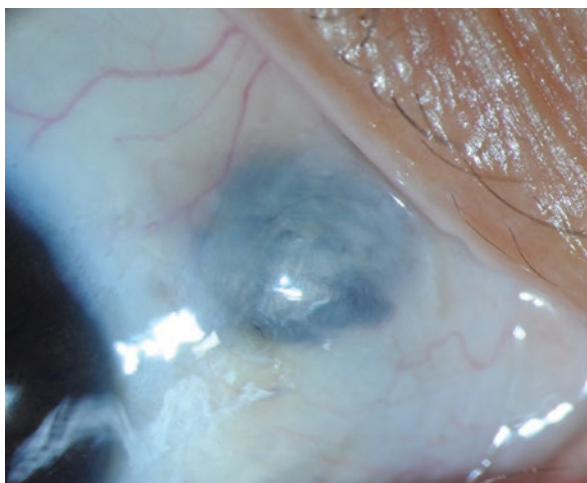
the COMS study, 83% of study eyes had reported some level of new cataract development, and 12% underwent cataract surgery [15]. Improvement in visual acuity was seen in the majority of patients that underwent cataract surgery, with patients that did not experience an improvement demonstrating other side effects of radiation such as radiation retinopathy.

Prevalence of cataract formation does appear to be dependent to some extent on radiation dose and tumor location. In the COMS follow-up, 18% of patients that received a radiation dose of ≥ 24 Gy to the lens underwent cataract surgery compared to only 4% of patients that received a radiation dose of < 12 Gy to the lens [15]. Several other studies have also demonstrated increased risk of cataract surgery in patients with increased radiation dose to the lens [25, 34, 49, 67].

Sclera

Side effects to the sclera are few and uncommon following radiation treatment. In rare cases, scleral necrosis may be observed, though most mild cases can be managed minimally with observation (Fig. 10.2) [10, 16, 35, 36, 68]. Severe cases of necrosis can require grafting or, in some cases, enucleation. Scleral necrosis is most commonly seen in cases of larger, more anterior tumors that require higher doses of radiation. In a retrospective review, Correa et al. reported that 8 of 87 patients (9.2%) treated with plaque brachytherapy for posterior uveal melanoma developed scleral necrosis during follow-up [17]. Four patients were successfully treated with a scleral graft, three were managed by observation, and one required enucleation following two unsuccessful scleral grafts. In a study by Gunduz et al., scleral necrosis was observed in 15 of 136 patients (11%) treated with plaque brachytherapy for

Fig. 10.2 Scleral necrosis in a patient treated with plaque brachytherapy for an anterior uveal melanoma



ciliary body melanoma [36]; tumor thickness and increased intraocular pressure were found to be risk factors for development of scleral necrosis. The authors theorized that thicker tumors may indicate increased risk of scleral invasion, which would explain consistency with previous studies which have shown that scleral necrosis generally develops in the treatment of tumors that have invaded the scleral lamellae [59].

Choroid

Following radiation treatment, the choriocapillaris may be damaged and lead to atrophy of the retinal pigment epithelium [1]. Over time, the larger choroidal vessels may become less perfused and lead to vascular closure. Choroidal neovascularization can occur due to increased VEGF levels from ischemia, but the new blood vessels are often abnormal and weak. Choroidal neovascularization can be treated as described by intravitreal anti-VEGF injections.

Optic Nerve

Radiation optic neuropathy is characterized by sudden, painless vision loss within the first several months to years after radiation therapy [18, 45, 52]. Radiation damage to the optic nerve can result in atrophy and pallor and may present as optic nerve edema. However, in some cases, the optic nerve may not appear clinically different if damage occurs in the retrobulbar space. Upon presentation with sudden vision loss, physicians must differentiate between optic neuropathy and tumor recurrence, though tumor recurrence typically presents with a slower loss of visual acuity [35]. Generally, prevalence of radiation-induced optic neuropathy is increased among patients treated with more than 50 Gy in total radiation dosage to the optic nerve head [83].

Treatment for radiation optic neuropathy is still debated with limited prospective literature. The use of intravitreal steroids may be useful in reducing edema, though visual gains may be small and transient [22, 69, 78]. Finger and Chin have also used intravitreal bevacizumab in a small case series of patients affected by radiation optic neuropathy with anatomic and visual improvements [28].

Conclusion

In the last few decades, plaque brachytherapy has become an increasingly popular treatment option for uveal melanoma and has allowed patients to keep their eye and maintain visual acuity. However, visual and anatomic side effects from radiation

damage are common in the months and years following therapy with limited options for follow-up treatment. In the COMS trial, radiation retinopathy, cataract development, and optic neuropathy were the most frequently observed side effects in patients treated with plaque brachytherapy, with 45% of patients demonstrating a visual acuity of 20/200 or worse by the 3-year time point. Thus, the benefits and drawbacks of radiation compared to enucleation must be carefully reviewed with patients prior to treatment, and risk of radiation damage must be assessed. Radiation retinopathy, one of the most common causes of visual loss following brachytherapy, currently has no FDA-approved treatment, though multiple prospective trials have demonstrated excellent visual and anatomic outcomes when patients are treated with intravitreal anti-VEGF injections. The optimal treatment regimen for radiation retinopathy is still debated, however, with no widely accepted management plan among vitreoretinal specialists. Additional trials are ongoing, which will hopefully provide more insight into the efficacy and optimal management strategy for patients with radiation retinopathy. Overall, plaque brachytherapy and other radiation therapies still provide an effective alternative to enucleation for uveal melanoma, though patients should be carefully followed by an ophthalmologist in the years after for the development of damage to any important ocular structures.

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